Ryan P. Seaman

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SUMMARY

Biomedical informatics researcher and MMSc candidate at Harvard Medical School with experience in collaborative data analysis, research visualization, and scientific training. Interested in research and project roles connecting biomedical science with useful health technology.

EXPERIENCE

Graduate Research Assistant, HIDIVE Lab Dec 2025 - Present
Harvard Medical School | Boston, MA

- Developing VitessceGen, a multi-agent AI tool that generates interactive visualization configurations from spatial and single-cell biomedical data.
- Evaluating generated configurations against expert-authored examples using HuBMAP datasets, under Nils Gehlenborg's supervision.

Summer Research Intern, HIDIVE Lab May 2026 - Aug 2026
Harvard Medical School | Boston, MA

- Created the full annotation and narrative system within the existing Vitessce framework, with supervision and guidance from Mark S. Keller and Nils Gehlenborg.
- Connected narrative text, saved views, and visual annotations into guided stories for spatial and single-cell data; manuscript in preparation.

Bioinformatician Jan 2023 - Aug 2025
MDI Biological Laboratory, Comparative Genomic and Data Science Core | Bar Harbor, ME

- Collaborated with research labs on biomedical data analysis and developed tools that help researchers explore their own datasets.
- Built scscape, an open-source Nextflow pipeline for reproducible, multi-sample single-cell RNA sequencing analysis.
- Co-taught a three-day single-cell analysis course and led Python training and mentoring for summer interns.
- Maintained GCP/AWS infrastructure for research analysis and data storage.

EDUCATION

Harvard Medical School Expected May 2027
MMSc, Biomedical Informatics | Boston, MA

Colby College Jan 2023
BA, Computational Biology | Waterville, ME | Varsity soccer, four years

SKILLS

Research: Single-cell and spatial genomics; RNA-seq; 10x Visium, Xenium, and Multiome; data visualization
Tools: Python, R, Bash, Nextflow, Git, GCP, AWS, Slurm
Collaboration: Scientific training, course development, teaching, and research mentoring

SELECTED PUBLICATIONS

Seaman RP, et al. (2024). A cloud-based training module for efficient de novo transcriptome assembly using Nextflow and Google Cloud. Briefings in Bioinformatics.

Paulmann A, et al. (2026). Sodium-glucose co-transporter 2 inhibition improves age-dependent kidney microvascular rarefaction. Kidney International.